

Machine, Data and Learning

Utility Theory + Decision Theory
Chapter 17 from the book by Russell
and Norvig

Decision Theory

(How to make decisions)

Decision Theory

= Probability theory + Utility Theory
(deals with chance) *(deals with outcomes)*

- *Fundamental idea:*
 - The **MEU** (Maximum expected utility) principle
 - Agent is **rational** if and only if it chooses the action that yields the highest expected utility, averaged over all possible outcomes of the action
 - Weigh the utility of each outcome by the probability that it occurs

Revisiting Romania example

- If plan1 and plan2 are the two plans:
 - Plan 1 uses route 1
 - $P(\text{home-early} | \text{plan1}) = .8$, while $P(\text{stuck1} | \text{plan1}) = .2$
 - Route 1 will be quick if flowing, but stuck for 1 hour if slow
 - $U(\text{home-early}) = 100$, $U(\text{stuck1}) = -1000$
 - Assigned numerical values to outcomes!
 - Plan 2 uses route 2
 - $P(\text{home-somewhat-early} | \text{plan2}) = .7$, $P(\text{stuck2} | \text{plan2}) = .3$
 - Route 2 will be somewhat quick if flowing, but not bad even if slow
 - $U(\text{home-somewhat-early}) = 50$, $U(\text{stuck2}) = -10$

Application of MEU Principle

- $$\begin{aligned} EU(\text{Plan1}) &= P(\text{home-early} \mid \text{plan1}) * U(\text{home-early}) \\ &\quad + P(\text{stuck1} \mid \text{plan1}) * U(\text{stuck1}) \\ &= 0.8 * 100 + 0.2 * -1000 = -120 \end{aligned}$$
- $$\begin{aligned} EU(\text{Plan2}) &= P(\text{home-somewhat-early} \mid \text{plan2}) * U(\text{home-somewhat-early}) \\ &\quad + P(\text{stuck2} \mid \text{plan2}) * U(\text{stuck2}) \\ &= 0.7 * 50 + 0.3 * -10 = 32 \end{aligned}$$

EU (plan2) is higher, so choose plan2

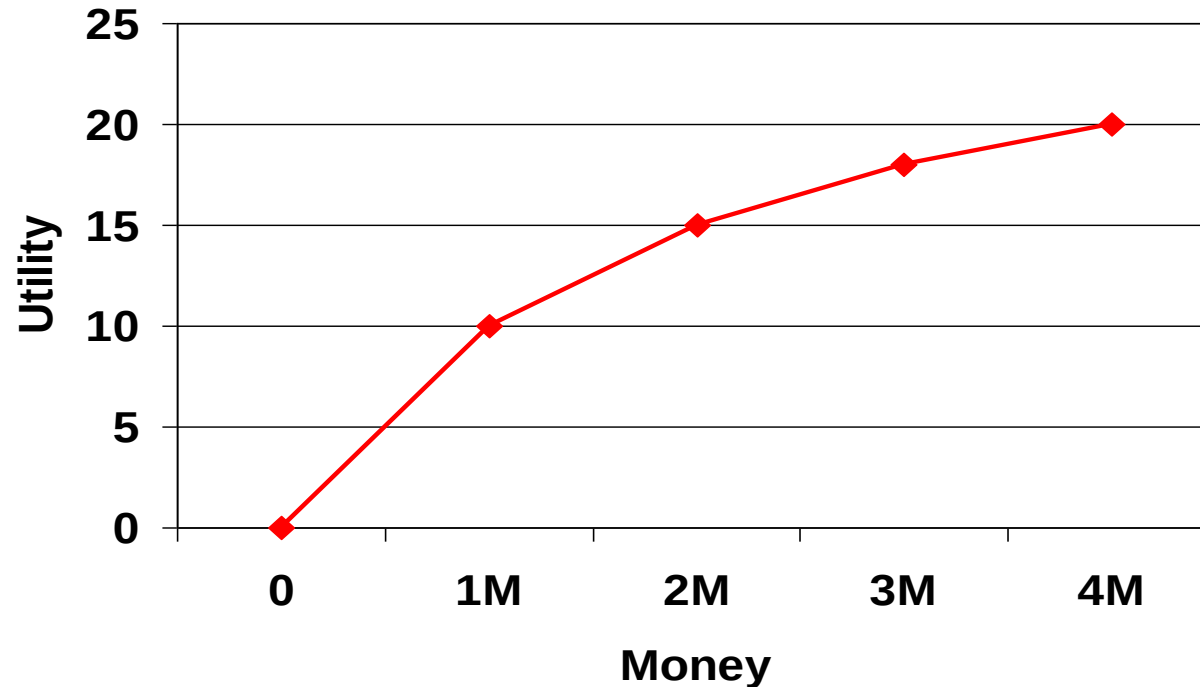
Lottery Example

- Suppose an agent gives you a choice:
 - **Choice 1:** You will get \$1,000,000
 - **Choice 2:** The agent will toss a coin
 - If heads, then you win \$3,000,000
 - If tails, then you get nothing
- Simple expected utility calculations give:
 - $EU(\text{Choice1}) = \$1,000,000$
 - $EU(\text{Choice2}) = \$1,500,000$
- So why did we prefer the first choice?

Risk Aversion

- We are **risk averse**
- Our utility functions for money are as follows (!!):
 - Our first million means a lot $U(\$1M) = 10$
 - Second million not so much $U(\$2M) = 15$ (NOT 20)
 - Third million even less so $U(\$3M) = 18$ (NOT 30)
 -
- Additional money is not buying us as much utility
- If we plot amount of money on the x-axis and utility on the y-axis, we get a concave curve

Answer: Risk Aversion



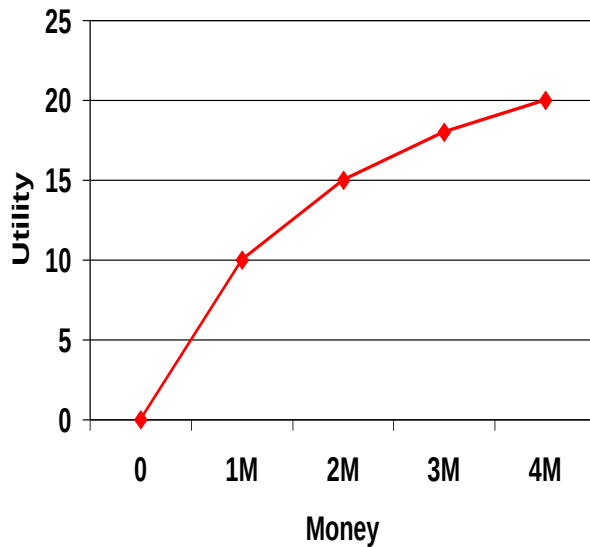
- $EU(\text{choice1}) = U(\$1\text{M}) = 10$
- $EU(\text{choice2}) = 0.5 * U(0) + 0.5 * U(\$3\text{M} = 18) = 9$
- That is why we prefer the sure \$1M

More Risk Aversion

- Key: Slope of utility function is **continuously decreasing**
 - We will refuse to play a monetarily fair bet
- Suppose we start with x dollars
 - We are offered a game:
 - 0.5 chance to win 1000 dollars ($c = 1000$)
 - 0.5 chance to lose 1000 dollars ($c = 1000$)
 - Expected monetary gain or loss is zero (hence monetarily fair)
 - Should be neutral to it, but seems we are not! Why?
 - $U(x + c) - U(x) < U(x) - U(x - c)$
 - $U(x + c) + U(x - c) < 2 U(x)$
 - $[U(x + c) + U(x - c) / 2] < U(x)$
 - $EU(\text{playing the game}) < EU(\text{not playing the game})$

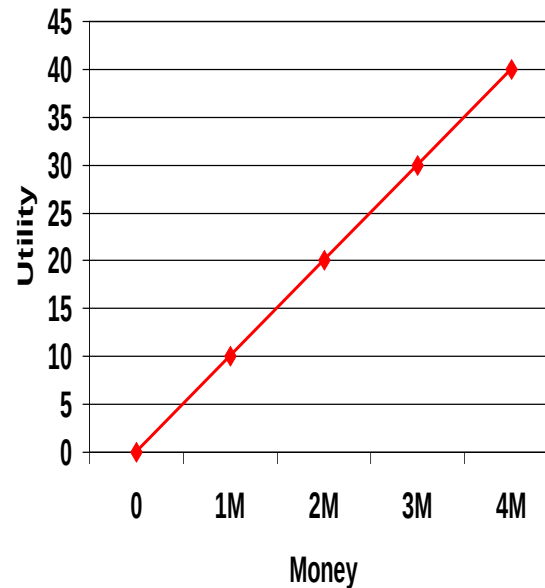
Risk Averse, Risk Neutral Risk Seeking

RISK AVERSE



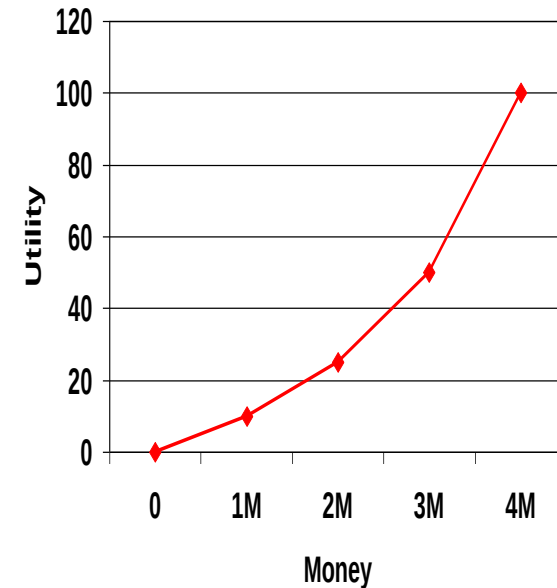
$EU(\text{Choice1}) = 10$
 $EU(\text{Choice2}) = 9$

RISK NEUTRAL



$EU(\text{Choice1}) = 10$
 $EU(\text{Choice2}) = 15$

RISK SEEKER



$EU(\text{Choice1}) = 10$
 $EU(\text{Choice2}) = 25$

Multiattribute Utility Theory

- Can we capture desirability of outcomes in a single utility function?
- Suppose renting an apartment
 - House1: closer-to-university, newer, costs 100 units
 - House2: Farther-from-university, older, costs 85 units
 - (Assume, you can afford up to 100 units)
- Outcomes characterized by two or more attributes
 - Attributes: $X_1, X_2, \dots, X_N$, e.g., <distance-to-univ, old/new, cost>
 - Values: $x_1, x_2, \dots, x_N$,
 - Closer-to-univ = 1, farther-from-univ = 0; new = 1, old = 0
 - Apartment1: <1,1,-100> Apartment2: <0,0, -85>
 - Which is a better apartment? (Pairwise comparison fails)

Multiattribute Utility Theory

- Don't get a single number, but vector of values as outcomes, $\langle x, y \rangle$
- How do you compare values now?
 - Compare $\langle 1, 1, -100 \rangle$ with $\langle 0, 0, -85 \rangle$
 - Compare $\langle 3, 3, 5 \rangle$ with $\langle 5, 3, 3 \rangle$
- One approach is dominance (strict, stochastic...):
 - If you are lucky, find $\langle 3, 3, 3 \rangle$ and $\langle 3, 3, 5 \rangle$
 - Values in one vector dominate values in the other vector

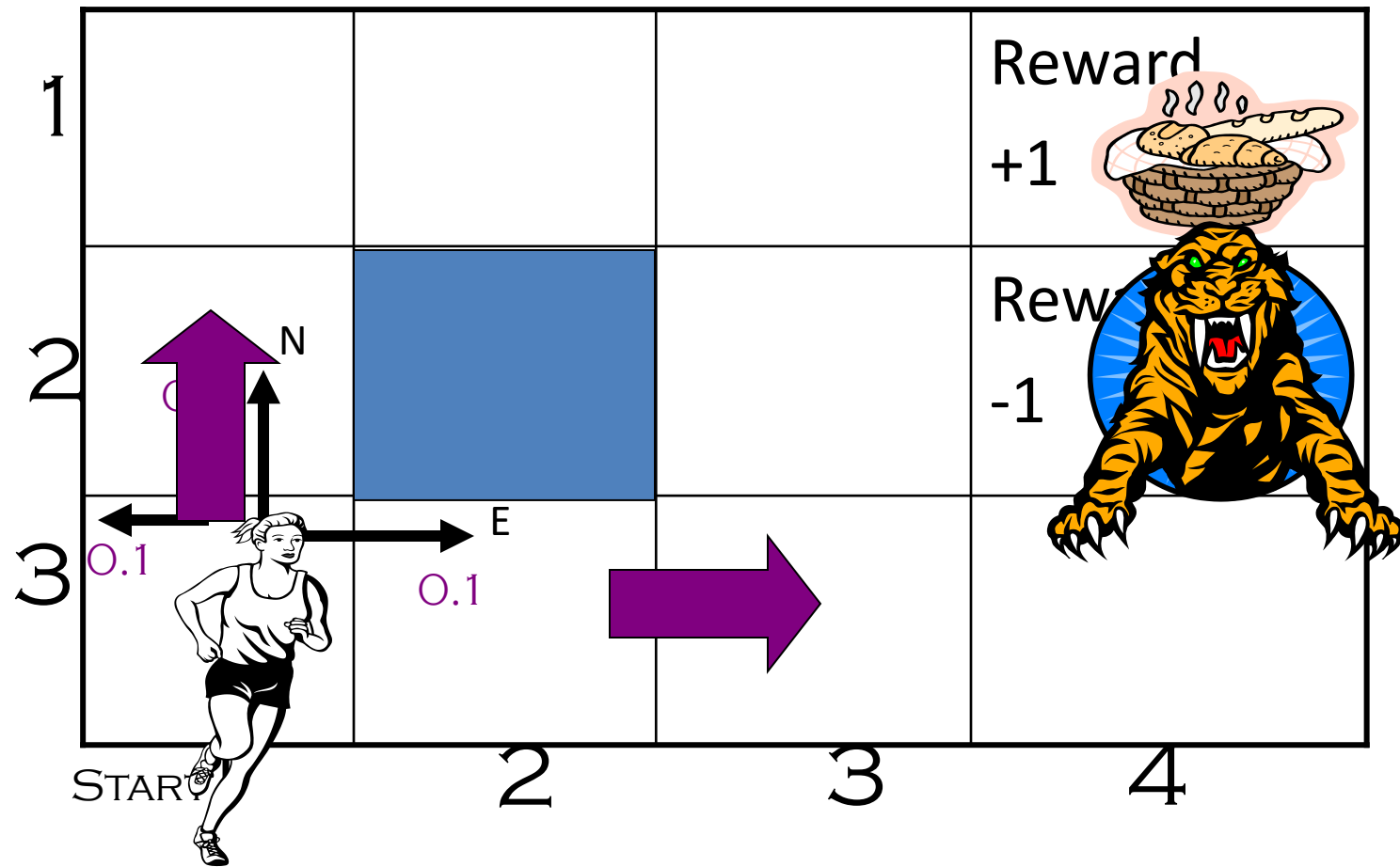
Markov Decision Process (MDP)

Chapter 17: Making Complex Decisions

- Defined as a tuple: $\langle S, A, P, R \rangle$
 - **S**: State
 - **A**: Action
 - **P**: Transition function
 - Table $P(s' | s, a)$, prob of s' given action “a” in state “s”
 - **R**: Reward
 - $R(s, a)$ = cost or reward of taking action a in state s
- Choose a sequence of actions (not just one action)
 - Utility based on a sequence of actions
 - Model **Sequential Decision Problems**

Example: What SEQUENCE of actions should our agent take?

- Agent can take action N, E, S, W
- Each action costs $-1/25$

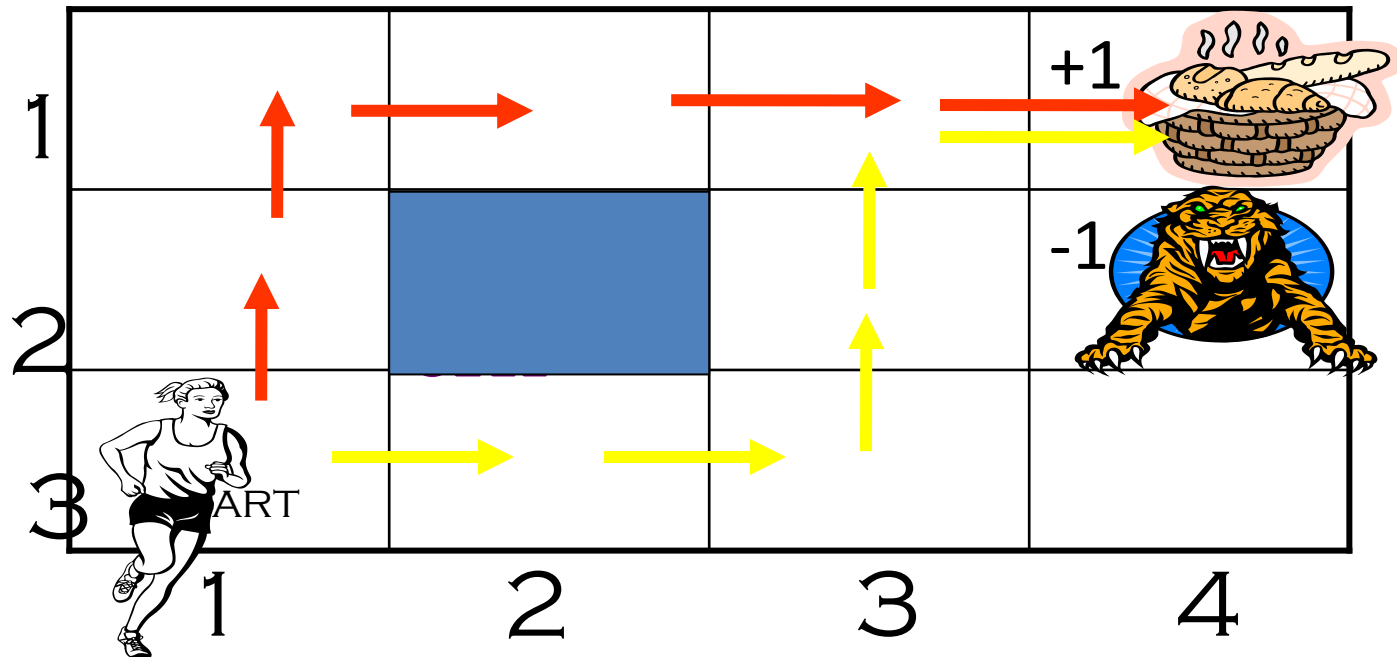


MDP Tuple: $\langle S, A, P, R \rangle$

- **S:** State of the agent on the grid
 - Ex: state (4,3) ([column, row] notation used)
- **A:** Actions of the agent, i.e., N, E, S, W
- **P:** Transition function
 - Table $P(s' | s, a)$, prob of s' given action “a” in state “s”
 - E.g., $P((4,3) | (3,3), N) = 0.1$
 - E.g., $P((3, 2) | (3,3), N) = 0.8$
 - (Robot movement, uncertainty of another agent’s actions,...)
- **R:** Reward
 - $R((3, 3), N) = -1/25$
 - $R(4,1) = +1$

How Would you Solve this Problem?

- Simple search algorithm? **Not deterministic**
- Apply MEU to an entire sequence of actions?
 - *Create multiple plans, e.g., Red plan vs. Yellow plan below*
 - *Choose a plan that leads to MEU*



How Would you Solve this Problem?

- Apply **MEU** to an entire sequence of actions?
- Does not work because uncertainty at every step
 - E.g., After first step of Red plan, may move east not North!
 - No action specified there (i.e., in cell (2,1))
- Solution is a *Policy*
 - Complete mapping from states to actions

MDP Basics and Terminology

- **Markov Assumption:** Transition probabilities (and rewards) from any given state depend only on the state and not on previous history
- An agent must make a decision or control a probabilistic system
 - Goal is to choose a sequence of actions for optimality
 - **Decision Epoch:** Points at which decisions are made
 - **Finite horizon MDPs:** # of decision epochs is finite i.e. fixed time after which game ends : Time dependent policy
 - **Infinite horizon MDPs:** # of decision epochs is infinite i.e. Time independent policy
 - **Transition model:** Table of probabilities P
 - In our example, 0.8, 0.1, 0.1 transition probabilities
 - $P(J \mid S, A)$: Probability of state J , given action A in State S
 - **Absorbing state:** Goal state

Reward Function

- Reward is assumed associated with state, action i.e. **$R(S, A)$**
 - If all actions have the same reward can use $R(S)$
 - We could also assume a mix of $R(S,A)$ and $R(S)$
 - Will use $R(S,A)$ as the notation
- Sometimes, reward associated with state, action, destination-state
 - $R(S,A,J)$
 - $R(S,A) = \sum J R(S,A,J) * P(J \mid S, A)$

MDP Policy

- **Decision Rule:** Procedure to choose action in each state for *a given decision epoch*
 - E.g., MDP has states, S1 and S2, with actions A1, A2 in both states
 - Decision rules D_i for each decision epoch “i” as shown in table below
 - Four decision rules shown, D1, D2, D3, D4, one for each epoch
 - Numbers in (..) are probabilities, e.g., 0.7, 0.3, 1.0
- **Policy:** Decision rule to be used at all decision epochs
 - Policy = {D1, D2, D3, D4} (assuming finite horizon $T = 4$)

D1	D2	D3	D4...
S1 → A1 (0.7)	S1 → A1 (1.0)	S1 → A2 (1.0)	
→ A2 (0.3)	S2 → A1 (0.3)	S2 → A2 (1.0)	
S2 → A2 (1.0)	→ A2 (0.7)		

Stationary and Deterministic Policies

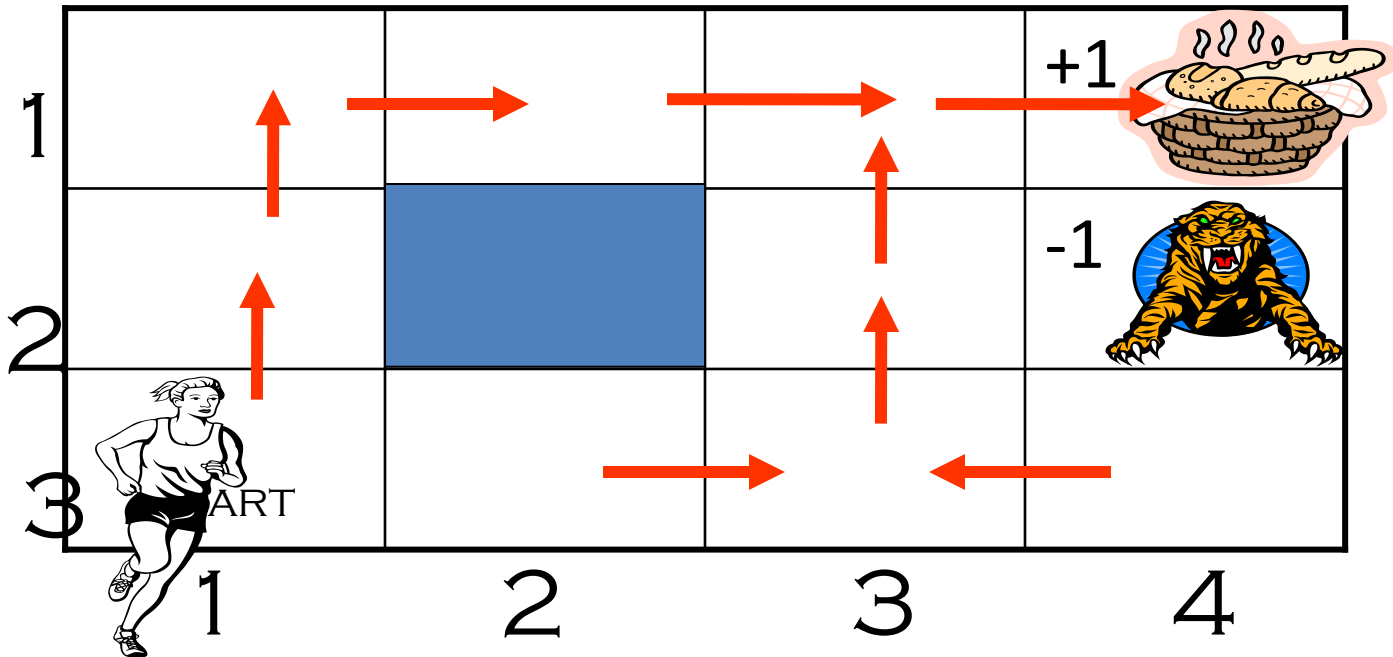
- **Stationary** policy implies same decision rule in every epoch
 - **Stationary policy:** {D, D, D, D...}
 - **Non-stationary policy** changes with time (e.g., D1,D2, D3...Dn)
- **Deterministic policy** implies choosing an action with certainty
 - **Deterministic policy:** $S_i \rightarrow A_i$ (probability 1.0)
 - **Randomized policy:** Probability distribution on the set of actions
- What type of a policy is the following?

D	D	D	D
$S1 \rightarrow A1$ (1.0)	$S1 \rightarrow A1$ (1.0)	$S1 \rightarrow A1$ (1.0)	$S1 \rightarrow A1$ (1.0)
$S2 \rightarrow A2$ (1.0)	$S2 \rightarrow A2$ (1.0)	$S2 \rightarrow A2$ (1.0)	$S2 \rightarrow A2$ (1.0)

Stationary and Deterministic Policies

- **Optimal** MDP policy for infinite horizon is Stationary & Deterministic policies (aka pure policy)
- Policy denoted by symbol π
- Stationary & deterministic policies denoted π^{SD}
- Is a policy π^{SR} possible? (SR = Stationary & randomized)
- **Note:**
 - When nothing is specified regarding time horizon, assume infinite horizon
 - When asked to find the policy **at** time horizon = 4, it means find the decision rule D4. It can also be stated as find decision rule for $T = 4$ or D4.
 - When asked to find policy for a time horizon of 4, means find all decision rules D1, D2, D3 and D4

Pure Policies: π^{SD}



- Deterministic, non-changing mapping from states to actions

$\pi((1,3)) \rightarrow \text{North}$ (non-changing, non-random)

$\pi((1,2)) \rightarrow \text{North}$

$\pi((4,3)) \rightarrow \text{West} \dots$

Policy

- **Policy** is like a plan, but not quite
 - Certainly, generated ahead of time, like a plan
- Unlike traditional plans, it is not a sequence of actions that an agent must execute
 - If there are failures in execution, agent can continue to execute a policy
- Prescribes an action for all the states
- Maximizes expected reward, rather than just reaching a goal state

Value Iteration: Algorithm

- *Basic algorithm is very simple!*

- *Initialize: $U_0(I) = 0$*

- *Iterate:*

$$U_{t+1}(I) = \max_A [R(I,A) + \sum_J P(J|I,A) * U_t(J)]$$

–Until close-enough (U_{t+1}, U_t)

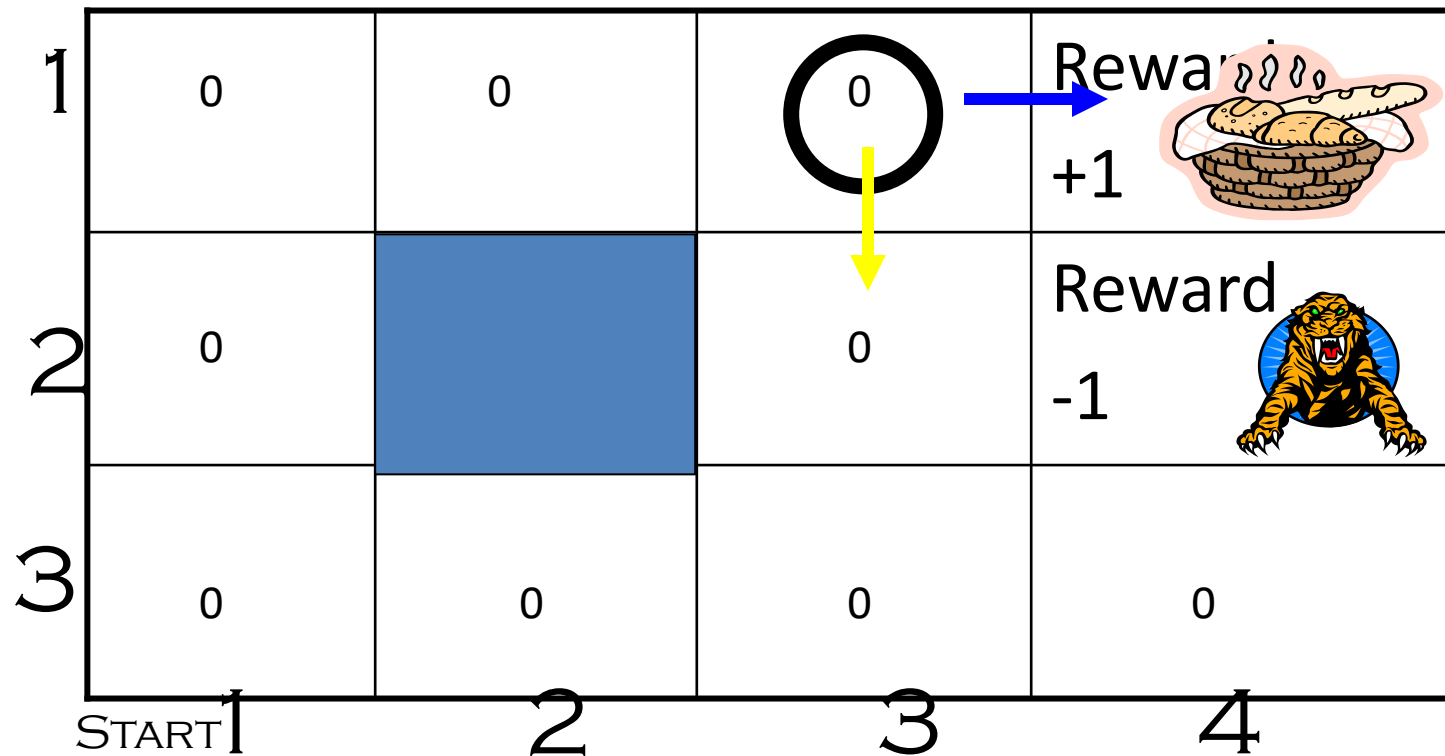
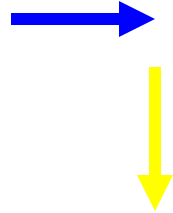
Dr. Richard Bellman

- *Iteration step called “Bellman update”*
- *Inventor of dynamic programming (1957)*



Iteration #1: Cell (3,1)

- East: $-1/25 + [0.8 * 1 + 0.1 * 0 + 0.1 * 0] = 0.76$
- North/South: $-1/25 + [0.8 * 0 + 0.1 * 1 + 0.1 * 0] = 0.06$
- West: ??
- So, State (3,1) has value of 0.76



Markov Chain

Discounting

Value Iteration: Modify

- *Initialize:* $U_0(I) = 0$

- *Iterate:*

$$U_{t+1}(I) = \max_A [R(I,A) + \gamma \sum_J P(J|I,A) * U_t(J)]$$

– Until close-enough (U_{t+1}, U_t)

- At the end of iteration, calculate optimal policy:

$$Policy(I) = \underset{A}{argmax} [R(I,A) + \gamma \sum_J P(J|I,A) * U_{t+1}(J)]$$